

Some Techniques Used for the Third Dimension

How do we perceive the third dimension, that is, how do we distinguish between two and three dimensions? The answer is here

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The nature has provided us with such a beautiful optical system that it has no match so far. Our eyes with which we see the world individually form two-dimensional (2D) pictures and the brain processes this data and provides us the depth perception.

One-eyed persons also perceive depth as they use the other biological tools provided by nature, though overall they have monocular vision. Tools that enable depth perception basically include:

1. **Binocular vision.** The *binocular-ity.org* states that our two eyes form images of a point that are slightly separated, denoted by 'd' (Fig. 2), and the brain converts those two different images to a single image in depth.

From this we can conclude that if we have smaller eye separation, depth perception will be more.

2. **Accommodation.** It is basically the ability to focus on close as well as distant objects. In doing so, the shape of our eyes changes and thus provides us a clue of the depth.

3. **Parallax.** It is basically the apparent difference in the position of an object when viewed from different positions.

4. **Size familiarity.** By this, our brain approximates the distance if it is aware of object size. For example, take two pencils of the same size and place them at different distances. The one looking bigger will be closer. But our brain can interpret their sizes correctly.

5. **Aerial perspective.** This is basically related to contrast. Contrast is simply the difference in colours

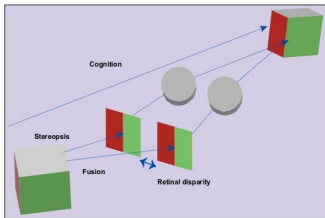


Fig. 1: Basic process showing how real-world objects are perceived. Stereopsis is the cognitive sensation and retinal disparity is the inter-ocular distance (Courtesy: www.affonsobeatto.com)

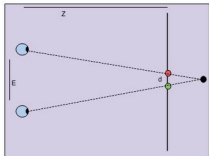


Fig. 2: Concept of binocular vision. E is inter-ocular distance, Z is viewing distance and d is physical disparity (Courtesy: www.binocularly.org)

and brightness within the same field of view. The human eyes are more sensitive to contrast than to luminance. As light is scattered by every object, a closer object would have a better contrast than a distant object.

These tools are used by our optical system simultaneously. However, due to limitation of technology, the 3D videos use only a combination of three or four.

Some eye problems and diseases prevent our eyes to perceive depth. These include, loss of vision in one eye,

amblyopia (loss of eyes' ability to see details), strabismus (when the two eyes do not line up in the same direction) and optic nerve hypoplasia (nerves not ending correctly).

The technology

In movies, the depth perception is achieved by colours, contrast, movement, perspective

art and spatial presentation. The first 3D film was presented at Astor Theatre, New York on June 10, 1915.

Earlier, methods used anaglyphs (wavelength-specific encoding) for stereoscopic 3D effect. The most common combination was red and cyan. The viewers were provided with eyeglasses having different wavelength-specific filters.

The filter colours have to be opposite on the VIBGYOR colour spectrum to avoid crosstalk between two independent images formed by the eyes. It was a convention to wear cyan (RGB:000,255,255) filter on the left eye and red (RGB:255,000,000) filter on the right. The filters allowed the wavelength opposite to the sheet colour to pass through them while the same wavelength was blocked. The anaglyphs basically used the stereoscopic vision that exploits the binocular disparity of our eyes.

Basic classification

3D content can be created with glasses